



Abstract ID : 1520

The frequency domain method of studying the Electric Dipole Moment in the quasi-frozen spin version of the NICA collider

Content

To search for the deuteron EDM, the development of a specialized ring corresponding to the “frozen spin” regime is commonly discussed. In this mode, the spin is always aligned with the momentum through the use of special deflectors with crossed electric and magnetic fields. To study the EDM, we must split out the influence of the EDM on spin precession and measure it. For this purpose, the Frequency Domain Method was developed, allowing the EDM contribution to the spin precession frequency relative to the radial axis to be greater than the contribution from the magnetic field, which arises from the inaccuracy of the ring element arrangement. Using the smallness of the deuteron magnetic anomaly, we developed the idea of “quasi-frozen spin,” where the magnetic and electric fields are separated and distributed into arcs and straight sections respectively. The deviation of the spin direction from the momentum in the arcs is compensated for by Wien filters on the straight sections. The simplest technical solution for the NICA collider, currently under commissioning, is the creation of bypasses at the collider equipped with Wien filters. However, the modulation of spin precession with Wien filters in NICA bypasses will change the contribution of the MDM to the overall spin precession frequency. In this paper, we analyze the frequency domain method in a quasi-frozen structure.

Footnotes

Funding Agency

the grant of the Russian Science Foundation No. 25-72-30005

In which format do you intend to submit your paper?

Word

Preprint marking on your proceeding paper

I no longer wish to present this contribution, please withdraw it.

Paper status

Author: Prof. SENICHEV, Yury (Russian Academy of Sciences)

Co-authors: AKSENTYEV, Alexander (Russian Academy of Sciences); Mr KOLOKOLCHIKOV, Sergey (Russian Academy of Sciences); MELNIKOV, Aleksei (Russian Academy of Sciences); PALAMARCHUKA, Patricija (Russian Academy of Sciences)

Presenter: Mr KOLOKOLCHIKOV, Sergey (Russian Academy of Sciences)

Track Classification: MC1.A30 - MC1.A30: Colliders: Polarized beams

Contribution Type: Poster Presentation

Submitted by **Prof. SENICHEV, Yury** <y.senichev@inr.ru> on **Tuesday, 2 December 2025**